

1. Applied Research > General > Model Evaluation
2. In my thesis on diffusion features for pose estimation, one of my ideas was to use a depth ControlNet as proxy for geometric understanding. However after initial experimentation I realized that the training objective of ControlNet merely refines the diffusion RGB features in the same latent space and depth info added which is necessary for correspondence tasks is marginal.
3. In AblnBev testing whether the new arch. actually learnt the intended effects was a huge problem and relied on domain experts' knowledge. I took the initiative to create a parametrized synthetic data generator where we simulate the varying levels of any effect on fake data to measure if the model is able to capture that and as fringe benefits we were able to integrate this into our CI/CD pipeline.
4. A few detrimental cases of accuracy hacking that come to my mind are: 1. Parallel to anomaly detection the model should be useful for predicting bad weather which might be uncommon but accuracy can be hacked by predicting more normal weather (gross simplification) 2. Weather prediction is a high uncertainty problem so accuracy gains without confidence calibration give less useful results.
5. My views on AI for coding is a bit regressive vs most people. I observed that coding agents create module abstractions often without a regard for future scaling. I have since switched to designing the classes with a lot of intervention at my end instead of full agent autonomy inside a testing loop. However my view will change soon given the rate of progress of frontier models.
6. I mostly follow alphaxiv for keeping track of latest research <https://x.com/askalphaxiv?s=20> but I also love Yann LeCun's takes on his twitter handle.